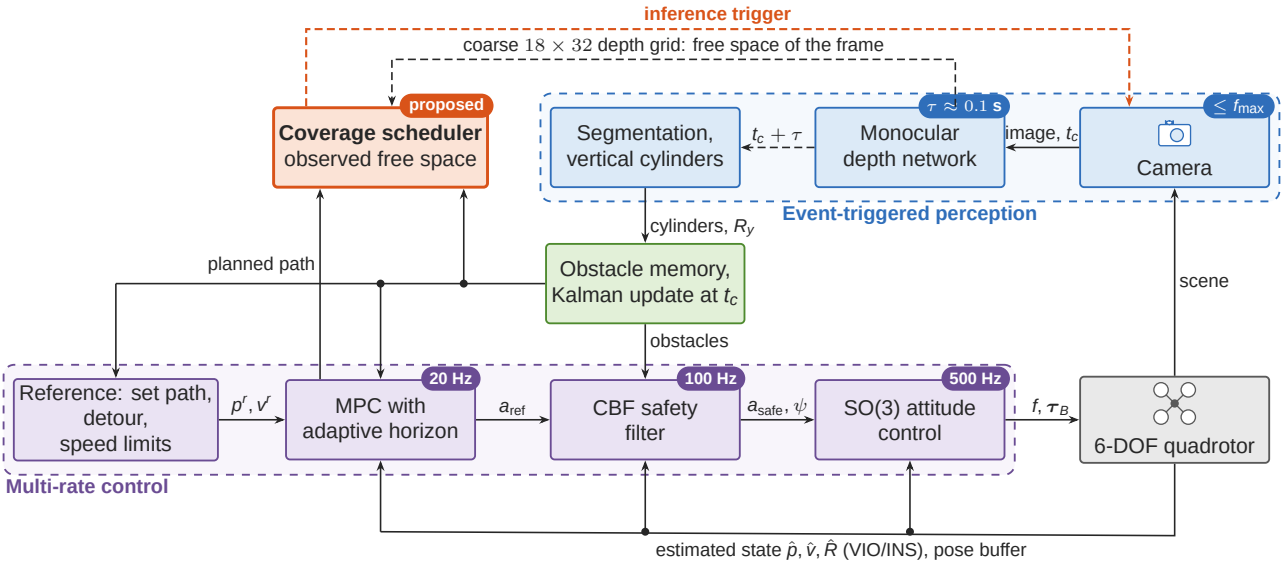
inference trigger
coarse 18×32 depth grid: free space of the frame
proposed
Coverage scheduler
observed free space
$\tau \approx 0.1$ s
$\leq f_{max}$
Segmentation, vertical cylinders
$t_c + \tau$
Monocular depth network
image, t_c
Camera
Event-triggered perception
cylinders, R_y
planned path
Obstacle memory, Kalman update at t_c
scene
obstacles
Reference: set path, detour, speed limits
p^r, v^r
20 Hz
MPC with adaptive horizon
a_{ref}
100 Hz
CBF safety filter
a_{safe}, ψ
500 Hz
SO(3) attitude control
f, τ_B
6-DOF quadrotor
Multi-rate control
estimated state $\hat{p}, \hat{v}, \hat{R}$ (VIO/INS), pose buffer